Group (Co)homology

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September 12, 2022

1 Introduction

Group homology started with the study of $K(G,1)$ spaces (a topological space X with $\pi_1(X) = G$ and $\pi_n(X) = 0$ for $n > 1$, or equivalently, with a simply connected universal cover). Hurewicz discovered that all such spaces are homotopy equivalent, so other topological properties of $K(G,1)$ spaces such as homology and cohomology, which are invariant under homotopy equivalence, can be considered properties of G. Further developments created purely algebraic methods of computing homology and cohomology in terms of G alone. Modifications to this algebraic machinery can also be used to determine information about the group actions of G. Further details on the subject beyond this paper can be found in [1].

2 Background

In this section, we motivate and introduce some of the algebraic tools that are used in the study of group homology theory, namely the group ring and free resolutions. As explained in the introduction, to define the homology of a group, we would like to start with a $K(G,1)$ space X and compute its homology. To do so, we make use of the associated simplicial chain complex $C(X)$. A closely related object is the chain complex $C(\tilde{X})$ of the universal cover $\tilde{X}$ of X. We can endow these chain complexes with more structure which will help us convert to a completely algebraic description of homology. As we will see, we can endow a G-action on X and $\tilde{X}$ which permutes its simplices and makes each of the chain groups into a module over a ring called the group ring $\mathbb{Z}G$ which is closely related to G. In algebraic topology, we are used to dealing with chain complexes of abelian groups, but this observation motivates us to generalize to chain complexes of R-modules for a given ring R. Of course, abelian groups are just $\mathbb{Z}$-modules, but this notion will allow us to consider $C(X)$ and $C(\tilde{X})$ as chain complexes of $\mathbb{Z}G$. The projection map $\tilde{X} \to X$ induces a map on the chain complexes which we will use to define homology. Moreover, since X is $K(G,1)$, $\tilde{X}$ is simply connected so the augmented chain complex has trivial homology groups for every degree. We will see that this translates to $C(\tilde{X})$ being a special type of chain complex called a free resolution. The power of free resolutions is any free resolution will give the same computation when we consider homology.

2.1 Chain Complexes

As stated, we want to consider the notion of chain complexes of R-modules for a given ring R so we start with a definition and some important properties. This notion is almost identical to a chain complex of abelian groups, except that boundary maps are R-linear homomorphisms of modules. Namely, a chain complex of R-modules consists of a graded R-module i.e. a sequence of R-modules $C = (C_n)$ indexed by the integers, and boundary maps $d_n : C_n \to C_{n-1}$ which are R-linear homomorphisms satisfying the chain condition $d_n \circ d_{n-1} = 0$ so that $\text{Im}(d_{n+1}) \subseteq \ker(d_n)$. The index n is called the degree of the module. The subscript indicating the degree of the module will often be suppressed with the index to be understood based on context, so for example, the chain condition will be written as $d^2 = 0$. Boundaries $B_n = \text{Im}(d_{n-1})$ and cycles $Z_n = \ker(d_n)$ are defined analogously to the abelian group case. We define the homology $H(C)$ of a chain complex to be B_n/Z_n. Exact sequences and cochain complexes, coboundaries, cohomology, etc. are defined analogously to the abelian group case. The remainder of the background will discuss chain complexes with the equivalent cochain ideas to be understood analogously.

A morphism called a chain map between complexes (C, d) and (C', d') is a sequence of homomorphisms $f = (f_n)$, $f_n : C_n \to C'_n$ compatible with the boundary map i.e. $fd = d'f$. Two chain maps f, g are homotopic if there is a sequence of maps $h_n : C_n \to C'_{n+1}$ called a chain homotopy such that $d'h + hd = f - g$. The motivation for this definition is that the existence of h implies f and g induce the same maps on homology.

Homotopic maps also allow us to have a weaker generalization of isomorphic chain complexes. Two chain complexes C, C' are said to be homotopy equivalent if there are morphisms $f : C \to C'$ and $f' : C' \to C$ such that ff' and $f'f$ are homotopic to the identity map of the respective chain complex. If two maps are homotopy equivalent, then they have the same homology.

2.2 The Group Ring $\mathbb{Z}G$ and G-modules

We now construct for each group G a ring $\mathbb{Z}G$ called the group ring. The elements of $\mathbb{Z}G$ are finite formal sums of the form $\sum x_i g_i$ for $x_i \in \mathbb{Z}$ and $g_i \in G$. Addition is defined in the obvious manner and multiplication is defined by distributing and evaluating the group products as follows: $(\sum_i x_i g_i)(\sum_j y_j g_j) = \sum_{i,j} x_i y_j (g_i * g_j)$ where $g_i * g_j$ is the element of G determined by the group multiplication.

Example 1. *If $G = \mathbb{Z}/2\mathbb{Z}$ with identity ι and nontrivial element τ, $\mathbb{Z}G$ consists of elements $a\iota + b\tau$ with multiplication $(a\iota + b\tau)(c\iota + d\tau) = (ac + bd)\iota + (ad + bc)\tau$.*

Example 2. *Generalizing the previous case, if $G = \mathbb{Z}/n\mathbb{Z}$, it can be easily verified that $\mathbb{Z}G \cong \mathbb{Z}[x]/(x^n)$.*

A $\mathbb{Z}G$-module, also called a G-module for brevity, is an abelian group A together with a homomorphism $G \to \text{Aut}(A)$, which induces a linear map $- * - : \mathbb{Z}G \times A \to A$. The map $G \to \text{Aut}(A)$ is just an action of G on A, so we can instead state that a G-module is a commutative group together with a specified G-action.

Example 3. *For any group G, we can consider $\mathbb{Z}$ as a G-module with the trivial group action. Explicitly, $ga \mapsto a, \forall g \in G, \forall a \in \mathbb{Z}$ and given $r \in \mathbb{Z}G$, $ra \mapsto \varepsilon(r)a$ where $\varepsilon : \mathbb{Z}G \to \mathbb{Z}$ is the augmentation map which takes $\sum x_i g_i$ to $\sum x_i$. The name augmentation map will become clear later when we consider the connection of this example to the topology of the $K(G,1)$ space.*

Example 4. *Let $G = \mathbb{Z}/2\mathbb{Z}$ with elements ι, τ. We can consider $\mathbb{Z}$ as a G-module with τ acting as multiplication by -1. Explicitly, $(a\iota + b\tau)c \mapsto ac - bc$ for $a, b, c \in \mathbb{Z}$.*

Example 5. *Given a set X that G acts on by permutations, we construct the group $\mathbb{Z}X$ of formal linear combinations of elements of X and extend the action of G linearly to make $\mathbb{Z}X$ (also denoted $\mathbb{Z}[X]$) into a G-module. As a particular example, given a subgroup $H \leq G$, G has a natural action on the left cosets G/H.*

However, orbit-stabilizer tells us this is essentially the general case. It can be easily checked that a disjoint union of sets corresponds to a direct sum of modules so $\mathbb{Z}\left[\coprod X_i\right] \cong \bigoplus \mathbb{Z}X_i$. From this, it can be easily checked that for any set X acted on by G, $\mathbb{Z}X \cong \bigoplus \mathbb{Z}G/G_x$, where G_x is the stabilizer group of x, as x ranges over a set of representatives for the orbits of X. In particular, if the action is a free action so that G_x is trivial for all x, $\mathbb{Z}X$ is a free $\mathbb{Z}G$-module.

Example 6. *Consider a field extension L/K with galois group $G = Gal(L/K)$. We can consider the additive group of E to be a G-module with G acting with the Galois action. We can similarly consider the multiplicative group $L^\times$ as a G-module again with the Galois action.*

2.3 Free Resolutions

Given an R-module M, a resolution is an exact sequence of R-modules $\ldots \to F_2 \xrightarrow{\partial_2} F_1 \xrightarrow{\partial_1} F_0 \xrightarrow{\varepsilon} M \to 0$. If all F_i are free, the resolution is said to be a free resolution. Constructing a free resolution is a simple procedure: we can consider M as a quotient of a free group F_0 by a subgroup of relations, giving a surjective map $\varepsilon : F_0 \to M$. We construct F_1 in the same way except replacing M with $\ker(\varepsilon)$ and proceed inductively in a similar fashion to construct all F_n. We denote such a free resolution as $F \to M$, but the actual maps of the resolution are only the F_i; the M is put for emphasis on the module.

Evidently, there is a lot of freedom in constructing a free resolution, which makes the following theorem surprising: any two free resolutions of a given module M are homotopy equivalent. We will not prove it here, but we can sketch the idea with the first chain map φ_1. The usefulness of the resolutions being free is that any homomorphism of free modules is determined by specifying where each generator goes. Let $\varepsilon, \varepsilon'$ be the maps $F_0, F_0' \to M$. Then, given a generator m of F_0, that maps to $\tilde{m} \in M$ arbitrarily choose some $m' \in F_0'$ which also maps to $\tilde{m}$ (obviously one exists since the map is surjective) and set $\varphi_1(m) = m'$. A map in the other direction φ_1' can be similarly constructed. The homotopy of $\varphi_1' \circ \varphi_1$ to the identity chain map is a bit more technical and won't be discussed.

3 Free Resolutions and Universal Covers

In this section, we explore how to obtain a free resolution for $\mathbb{Z}$, which is considered for the rest of this section as a G-module with trivial action, using the topology of $K(G,1)$ spaces. This is the beginning of showing how we can translate topological ideas into purely algebraic ones.

Consider a regular covering space $p : \tilde{X} \to X$ of a CW-complex X with G as the group of deck transformations. $\tilde{X}$ inherits a cell complex structure (the interior of a cell on $\tilde{X}$ is a connected preimage of the interior of a cell on X) and G acts on this cell complex under deck transformations. By linearity, this turns each of the chain groups $C_n(X)$ into G-modules. Moreover, G acts freely on the cells of $\tilde{X}$ i.e. $g\sigma \neq \sigma$ for $g \neq e$. As explained earlier, this makes the chain groups into free G-modules. Obviously, the action of G commutes with the boundary map of the chain complex $C(\tilde{X})$, but this means $C(\tilde{X})$ is not only a chain complex of abelian groups, but a chain complex of G-modules.

Now, we consider the case that X is a $K(G,1)$ space. In this case, the universal cover is a simply connected space with G as its group of deck transformations. $C(\tilde{X})$ is then a complex of G-modules, but since $\tilde{X}$ is simply connected, it is homotopy equivalent to a point so that all homology groups of $\tilde{X}$ except H_0 are trivial. If we instead consider the reduced chain complex, we have that all reduced homology groups are trivial, or in other words, the reduced chain complex is an exact sequence of G-modules (again treating $\mathbb{Z}$ as a module with trivial action). Finally, since all the chain groups are free, we see the chain complex ... $\xrightarrow{\partial} C_1(X) \xrightarrow{\partial} C_0(X) \xrightarrow{\varepsilon} \mathbb{Z} \to 0$ forms a free resolution of G-modules for the module $\mathbb{Z}$.

3.1 Standard Resolution

We can construct a $K(G,1)$ cell complex in a uniform way for any group G, and doing so will give us a standard free resolution of $\mathbb{Z}$ for any group. The complex is constructed by letting the vertices of X be labeled by the group elements of G, and making an ordered n-simplex for each $n+1$-tuple $(g_0, ..., g_n)$ with those elements as vertices. The constructed space is a universal over of a $K(G,1)$ space, and has the advantage of being able to compute the boundary maps of the chain complex explicitly in terms of the group elements.

Explicitly, the chain groups $C_n(X)$ are formal linear combinations of $n+1$-tuples $(g_0, ..., g_n)$ corresponding to the simplices. The boundary map is given by $\partial(g_0, ..., g_n) = \sum_i (-1)^i (g_0, ..., \hat{g}_i, ..., g_n)$. The G-action is simply $g(g_0, ..., g_n) = (gg_0, ..., gg_n)$ and is a free action. This makes $C_n(X)$ into a free G-module. Here, $C_0(X)$ consists of formal sums of the vertices, which are the elements of G. In other words, $C_0(X) \cong \mathbb{Z}G$ and the topological augmentation map ε takes any g to 1 which explains the name of the algebraic augmentation map.

We can find a canonical basis for this module in terms of group elements to try to simplify calculations. To find a representative for $(g_0, ..., g_n)$, we can multiply by g_0^{-1} to make the first vertex the identity e. It is often convenient to rewrite such a tuple as $(e, g_1, g_1g_2, ..., g_1g_2...g_n)$ so that each vertex is the previous multiplied by the appropriate group element. To make things cleaner, we use the bar notation: $[g_1|g_2|...|g_n] := (e, g_1, g_1g_2, ..., g_1g_2...g_n)$ and write the identity vertex as the empty bracket

$[\,] := (e)$. The boundary map on these generators is given by $\partial = \sum_i (-1)^i d_i$ with d_i defined as follows:

$$d_i = \begin{cases} g_1[g_2|...|g_n] & i = 0 \\ [g_1|...|g_{i-1}|g_i g_{i+1}|g_{i+2}|...|g_n] & 0 < i < n \\ [g_1|...|g_{n-1}] & i = n \end{cases}.$$

This isn't anything new and follows from our boundary map on the simplices, just rephrased into the bar notation. This resolution is often called the bar resolution of $\mathbb{Z}$.

In low dimensions, the maps are $C_2(X) \overset{\partial_2}{\to} C_1(X) \overset{\partial_1}{\to} C_0(X) \overset{\varepsilon}{\to} \mathbb{Z} \to 0$ given explicitly as $\partial_2([g_1|g_2]) = g[h] - [gh] + [g]$, $\partial_1([g]) = g[\,] - [\,] = (g - e)$, $\varepsilon([\,]) = \varepsilon((e)) = 1$.

4 Group Homology

We finally define the homology groups $H_n(G)$. As stated earlier, if we let Y denote a $K(G,1)$ space, we define $H_n(G)$ to be the topological homology groups $H_n(Y)$. If the description is so straightforward, why did we bother with all the algebra so far? As we now describe, we can formulate the homology groups completely algebraically. We said $H_n(G)$ should be a property of G alone, and with this, we will have a systematic method to compute the homology groups without ever needing to appeal to the topology.

The first thing to note is that for a universal cover X of the $K(G,1)$ space Y, we have associated maps $p : C_n(X) \to C_n(Y)$. The reason this is useful is because we have an algebraic description of $C_n(X)$: it is a free $\mathbb{Z}G$ resolution of $\mathbb{Z}$, and we know that any two such resolutions are homotopy equivalent. Hence, if we can figure out how to determine $C_n(Y)$ from $C_n(X)$, we can start with any free resolution of $\mathbb{Z}$ instead of $C_n(X)$ and use it to determine the homology groups.

$C_n(X)$ is generated by tuples $(g_0, ..., g_n)$, and since for any simplex $\sigma \subseteq Y$, G permutes the preimages of σ, $g(g_0, ..., g_n)$ and $(g_0, ..., g_n)$ are mapped to the same element of $C_n(Y)$ for all g. Hence, if we give $C_n(Y)$ a G-module structure with G acting trivially, we see that $p : C_n(X) \to C_n(Y)$ is a surjective G-module homomorphism. Since $p(g\sigma) = p(\sigma)$ for $\sigma \in C_n(X)$, $g\sigma - \sigma$ is in $\ker(p)$ so we get an induced map $p' : C_n(X)/M' \to C_n(Y)$ where M' is the submodule generated by all elements of the form $g\sigma - \sigma$. $C_n(X)/M'$ has one generator per simplex of Y, but so does $C_n(Y)$, so p' sets up an isomorphism of these modules. Hence, $C_n(Y)$ is created from $C_n(X)$ by "losing information of the action of G." To make this precise, and to create a general framework to work with algebraically, we discuss the concept of co-invariants.

4.1 Co-invariants of a Module

Definition 1 (Module of Co-invariants). *Given a G-module M, we define the module of co-invariants M_G to be the quotient module M/M' where M' is the submodule generated by $gm - m$ for $g \in G$, $m \in M$.*

The module of co-invariants is obtained by quotienting out by the action of G and is the largest quotient module of M on which G acts trivially. The name comes from its similarity to the module of G invariants M^G which is defined as the largest submodule of M on which G acts trivially. For example, if G is finite, M^G is generated by elements of the form $\sum_{g \in G} gm$.

We see that $C_n(Y)$ is the module of co-invariants to $C_n(X)$, which is a purely algebraic relationship of the chain groups that does not rely on topology. There is another method to construct the module of co-invariants by using the tensor product. Since G is not commutative, the tensor product $N \otimes_{\mathbb{Z}G} M$is technically only defined for a right action on N and a left action on M. However, we can turn a left action on N into a right action by declaring $gn = ng^{-1}$, so that the multiplication is in the right order. Now, $gn \otimes m = ng^{-1} \otimes m = n \otimes g^{-1}m$, which in essense says that $gn \otimes gm = n \otimes m$.

Now, as usual, let $\mathbb{Z}$ be a G-module with trivial action. In this case, the right $\mathbb{Z}$ action is also trivial. Then, $M_G \cong \mathbb{Z} \otimes_{\mathbb{Z}G} M$. To show this, first $1 \otimes_g m = 1g \otimes m = 1 \otimes m$. Hence, we have a well defined map $\mathbb{M}_G \to \mathbb{Z} \otimes_{\mathbb{Z}G} M$ sending $\bar{m} \to 1 \otimes m$. There is a map $\mathbb{Z} \otimes_{\mathbb{Z}G} M \to M_G$ as guaranteed by the universal property of tensor products given by $a \otimes m \to a\bar{m}$. These maps can be verified to be inverses so that we have the desired isomorphism. The advantage of this viewpoint is that we see the functor of G-invariants sending M to M_G is an additive functor since it is the tensor product of M with some module. The important conclusion is that it preserves the maps involved in chain homotopies and hence, preserves homotopy equivalences. What this means is that if we start with two free resolutions $F \to M$, $F' \to M$ and then take the G-invariants of each module and look at the induced maps of the chain complexes, we get two new chain complexes $F_G \to M_G$, $F'_G \to M_G$. F and F' are homotopy equivalent since they are free resolutions and hence F_G and F'_G must also be homotopy equivalent by the induced maps.

To summarize, recall that $C_n(Y)$ was the chain complex we used to define $H_n(Y)$ and it was the co-invariants of $C_n(X)$, a free resolution of $\mathbb{Z}$. We can now instead start with an arbitrary free resolution $F \to \mathbb{Z}$ over $\mathbb{Z}G$ and look at the induced chain complex of co-invariants F_G. By the above remarks, have that F_G is homotopy equivalent to $C_n(Y)$ and hence, we get the same homology groups. We have finally constructed a purely algebraic way to obtain the homology of a group G as we had set out to do.

4.2 Some Theorems on Homology

We can compute some of the low dimensional Homology by looking at the standard resolution. The last few terms of the standard resolution is $C_2(X) \overset{\partial_2}{\to} C_1(X) \overset{\partial_1}{\to} C_0(X) = \mathbb{Z}G$. Taking G-invariants, we get maps $C'_2(X) \overset{\partial_2}{\to} C'_1(X) \overset{\partial_1}{\to} C'_0(X)$ with $C'_n(X)$ being the module of co-invariants. We now use the bar notation to represent the images of the corresponding simplices, which are now G-invariant, and can consider the induced boundary maps: $\partial_2([g|h]) = g[h] - [gh] + [g] = [h] - [gh] + [g]$, $\partial_1([g]) = g[\ ] - [\ ] = [\ ] - [\ ] = 0$. In addition, since $C_0 \cong \mathbb{Z}G$, $C'_0 \cong \mathbb{Z}$. It follows that $H_0(G) \cong \mathbb{Z}$, and $H_1(G)$ is formed by linear combinations of $[g]$ modded out by $[g] + [h] - [gh]$ i.e. $H_1(G)$ is the abelianization of G. It is more technical to show, but $H_2(G)$ also has a uniform description for any group: choose a presentation $G = F/R$ where F is a free group. Then, $H_2(G) \cong (R \cap [F, F])/[F, R]$, where $[G, H]$ denotes the subgroup of all commutators of elements in G and H.

There is also an interesting analog of the Mayer-Vietoris sequence. If we have topological spaces

$X = X_1 \cup X_2$ with the intersection $Y = X_1 \cap X_2$ nonempty and connected, Van-Kampen tells us $\pi_1(X) = \pi_1(X_1) *_{\pi_1(Y)} \pi_1(X_2)$. We can ask the reverse question that if we have an amalgamated product of groups i.e. a group G fitting into the commutative diagram

$$\begin{tikzcd} A \arrow[r, "\alpha_2"] \arrow[d, "\alpha_1"'] & G_2 \arrow[d, "\beta_2"] \\ G_1 \arrow[r, "\beta_1"'] & G \end{tikzcd}$$

with $\beta_1 \circ \alpha_1 = \beta_2 \circ \alpha_2$ that satisfies a universal property, can we find associated spaces satisfying the Van-Kampen conditions that produce this amalgamated product. It turns out that if the α_i are injective so that A can be regarded as a subgroup of G_i, we can construct maps satisfying

$$\begin{tikzcd} Y \arrow[r, hook] \arrow[d, hook] & X_2 \arrow[d, hook] \\ X_1 \arrow[r, hook] & X \end{tikzcd}$$

where each space is the $K(-,1)$ space of the corresponding group in the previous diagram. The idea is that if we have an injection of groups, we get an injection of corresponding $K(-,1)$ spaces which gives us the maps $Y \hookrightarrow X_i$, then gluing $X_1 \coprod X_2$ along Y, it can be shown the quotient is a $K(-,1)$ space, and by Van-Kampen, the fundamental group is the desired one.

From the Mayer-Vietoris sequence on the spaces Y, X_i, X, we get an associated Mayer-Vietoris sequence of group homology ... $\to H_nA \to H_nG_1 \oplus H_nG_2 \to H_nG \to H_{n-1}A \to$ ... if the groups satisfy the amalgamation commutative diagram and A is injective in both maps.

5 Group Cohomology

Analogous to the topological setting, we can construct cohomology groups as well. Usually, we take homomorphisms from $\mathrm{Hom}(C(X), \mathbb{Z})$, but we can make a few modifications to convert to our group setting. First, we have been working with free resolutions, so we can replace $C(X)$ with a free resolution of $\mathbb{Z}$. Secondly, analogous to how we can do cohomology over coefficients, we can replace $\mathbb{Z}$ with some G-module M. Since both F_i and M are G-modules, instead of considering only homomorphisms of the abelian group structure, we should consider G-module homomorphims. We now have a chain complex: given a $\mathbb{Z}G$ resolution $F \to \mathbb{Z}$ and a module M, we define F^i to be $\mathrm{Hom}_{\mathbb{Z}G}(F_i, M)$ with cochain boundary maps induced from the chain boundary maps. The cohomology is defined as the homology of this cochain complex. Similar to how the tensor product construction gave us a homotopy equivalence of co-invariants, the Hom functor behaves nicely and gives us homotopy equivalences of cochain complexes, showing that the cohomology is independent of choices of resolution.

It should be noted that homology can also be done with coefficients in arbitrary G-modules, we simply replace the $\mathbb{Z}$ in the tensor product with some other G-module (with a right action as discussed in that section). The cohomology is also a natural consequence of trying to generalize the topology of a $K(G,1)$ space to an algebraic calculation. The details are omitted, but are similar to the homology case.

Example 7. *Suppose G is infinite cyclic with generator t. We have a free resolution of $\mathbb{Z}$ given by $0 \to \mathbb{Z}G \overset{t-1}{\to} \mathbb{Z}G \to \mathbb{Z} \to 0$. Given a module M, we consider the maps $Hom_G(F_n, M)$. Since both modules are $\mathbb{Z}G$ and we have the well-known fact that for any ring $Hom_R(R, M) \cong M$, the cochain complex looks like $M \overset{t-1}{\to} M \to 0 \to ....$ $H^0(G, M)$ consists of all $m \in M$ such that $(t-1)m = 0$ i.e. m is invariant under the action of G so that $H^0(G, M) = M^G$, the group of G-invariants mentioned earlier.*

It can be checked that $H^1(G, M) = M_G$. If we were to go through and calculate the homology groups, we would find $H^1 \cong H_0$ and $H^0 \cong H_1$, giving us the Poincaré duality of the $K(G,1)$ space S^1, but from a purely algebraic calculation.

6 Hilbert's Theorem 90

Hilbert's Theorem 90 is an application of group cohomology to number theory. While the theorem can be proved through classical methods, the group cohomology gives a nice viewpoint. The theorem concerns the kernel of the norm map of a field extension. In particular, if we have a field extension L/K with galois group $G = \mathrm{Gal}(L/K)$, we can consider the norm map $N(\alpha) = \prod_{\sigma \in G} \sigma(\alpha) \in \mathbb{Q}$. The norm map is a group homomorphism $L^\times \to \mathbb{Q}^\times$ so we can look at its kernel and look for the values for which $N(\alpha) = 1$. It turns out that if G is cyclic generated by σ_0 (e.g. if L is cyclotomic), then if $N(\alpha) = 1$, there is some $\beta \in L$ such that $\alpha = \sigma_0(\beta)/\beta$, which gives us some control over the kernel. One potential application is to find the units in the ring of integers $\mathcal{O}_L$. The norm map maps the ring of integers to the rational integers, so if $ab = 1$ in $\mathcal{O}_L$, we have that $N(a)N(b) = 1$ implying $N(a) = \pm 1$.

We prove this by showing that it is a consequence of the fact that $H^1(G, L^\times)$ is trivial. We want to look at homomorphisms from our chain groups of the standard bar resolution to $L^\times$. Let's start with C_0. C_0 is generated by [ ] so all maps are determined uniquely by where this generator goes. Hence, each map in C^0 is determined by a single element of $L^\times$ which is the image of [ ]. C_1 is generated by elements of the form $[\sigma]$ and C^1 is determined by where each generator is sent. Hence, C^1 is determined by any map $G \to L^\times$. It is important to note that these maps are arbitrary and need not be homomorphims since $[\sigma]$ and $[\tau]$ are independent basis elements over $\mathbb{Z}G$. Finally, C_2 is generated by elements of the form $[\sigma|\tau]$ with σ, τ arbitrary so that C^2 is determined by arbitrary maps $G \times G \to L^\times$.

To compute H^1, we need to compute the coboundary maps. Let's look at the image of the map $\partial^0 : C^0 \to C^1$ i.e. the 1-coboundaries. We know that $f \in C^0$ is determined by $f([\,]) = \beta$. Then, the image is

$$g([\sigma]) = \partial^0 f([\sigma]) = f(\partial_0[\sigma]) = f(\sigma[\,] - [\,]) = \sigma(f([\,]))/f([\,]) = \sigma(\beta)/\beta.$$

The second to last equality follows from the fact that f is a G-module homomorphism. Note that this is almost the term we want from the theorem. We must also compute $\ker(\partial^1)$.

$$\partial^1 f([\sigma|\tau]) = f(\partial_1[\sigma|\tau]) = f(\sigma[\tau] - [\sigma\tau] + [\sigma]) = \sigma(f([\tau]))f([\sigma])/f([\sigma\tau]) = 1$$

so that any map $G \to L^\times$ satisfying $f([\sigma\tau]) = \sigma(f([\tau]))f([\sigma])$ determines a 1-cocycle.

To complete the proof, we need a theorem about fields. For distinct automorphisms $\sigma_1, ..., \sigma_n$ of L, the σ_i are linearly independent. In other words, if $\sum a_i\sigma_i$ is the 0 homomorphism, all of the a_i are 0. The proof is omitted, but it is a simple linear algebra argument. Now, suppose we have a 1-cocycle f and consider the homomorphism $\sum_{\tau\in G} f(\tau)\tau : L \to L$. By the independence of the automorphisms, there is some $\theta \in L$ such that $\gamma = \sum_{\tau\in G} f(\tau)\tau(\theta) \neq 0$. Then,

$$\sigma(\gamma) = \sum_{\tau\in G} \sigma(f([\tau]))\sigma\tau(\theta) = \sum_{\tau\in G}(f([\sigma]))^{-1}f([\sigma\tau])\sigma\tau(\theta) = (f([\sigma]))^{-1}\sum_{\tau\in G} f([\sigma\tau])\sigma\tau(\theta) = (f([\sigma]))^{-1}\gamma$$

The last equality holds since $\sigma\tau$ goes over all elements of G as τ varies. Hence, we see that $f([\sigma]) = \sigma(\beta)/\beta$ for $\beta = \gamma^{-1}$, but this is exactly a coboundary, so we have shown $H^1(G, L^\times)$ is trivial.

Finally, we can prove Hilbert's theorem 90. We now restrict ourselves to the case that G is cyclically generated by σ_0 of order n. Suppose we have an element $\alpha \in L$ with $N(\alpha) = 1$. We want to construct a cocycle f for which $f(\sigma_0) = \alpha$. We can then show by induction that $f(\sigma_0^k) = \alpha\sigma_0(\alpha)...\sigma_0^k(\alpha)$ using the cocycle condition:

$$f([\sigma_0^k]) = f([\sigma_0^{k-1}\sigma_0]) = \sigma_0^{k-1}(f([\sigma_0]))f([\sigma_0^{k-1}]) = \sigma_0^k(\alpha)(\alpha...\sigma_0^{k-1}(\alpha)).$$

We see that $f([\sigma_0^n]) = N(\alpha) = 1 = f([e])$ so that this map is well defined. Now, f is a 1-cocycle, but since the first homology is trivial, it is also a coboundary. Hence, there is an element β such that $f([\sigma]) = \sigma(\beta)/\beta$. In particular, $\alpha = f([\sigma_0]) = \sigma_0(\beta)/\beta$ as desired. There is a similar additive version of the theorem: the trace is defined as $\mathrm{Tr}(\alpha) = \sum_{\sigma\in G}\sigma(\alpha)$ and we have that if $\mathrm{Tr}(\alpha) = 0$ and G is cyclic, then there is some $\beta \in L$ such that $\alpha = \sigma_0(\beta) - \beta$. The proof follows in a similar fashion.

Note that we never used the assumption that G is cyclic to show the first cohomology group is trivial. In fact, the first cohomology is always trivial for the Galois group acting on a field extension. This fact can be considered as a generalization of the Hilbert's theorem 90.

References

[1] Brown, Kenneth (1982). *Cohomology of Groups*. Springer Science & Business Media.